I chose the course planner project from my CS300 course. Originally this was done using C++ to load course information from a file and stores the courses that could be viewed in a list or be searched for. The program also shows all the prerequisites for all the courses. I chose this one for the enhancement because it already used data structures to organize and search for course information so it made sense and was also a good project to improve upon.

I chose this artifact because it shows I know how to work with data structures and algorithms to organize information. The original stored course information in a map and then allowed someone to load, search, or display the data. I wanted the program to do more between the courses instead of only storing and displaying them so I added a graph that shows the connections between the courses and the prerequisites. I also added validation to make sure the prerequisites listed actually existed in the course data. One of the main things I did to improve it was adding a DFS algorithm that checks for circular prerequisite connections so if one course required another then that eventually leads back to the first then the program will detect the cycle and tell the user the course plan isn't valid. I also added topological sorting so there was a recommended order in how the courses should be taken. This makes it so the prerequisite courses appear before the courses that need them. Both of those algorithms have a time complexity of O(V+E) based on the number of courses and prerequisites. I think making these changes helped improve the project because now the program can analyze the courses and their connections instead of just displaying the information that's loaded.

For this artifact I did meet the outcomes I originally planned. Outcomes 3 and 4 still fit for this because I improved the program by adding new ways to check for course information and sorting the courses in order that makes sense based on their prerequisites. I also got more experience working with algorithms and figuring out how to use them to improve something that was created before. I don't think a change to my plan is necessary because the enhancement still covers all the same skills I planned on showing.

Enhancing this artifact helped me understand how graphs can be used to solve problems with connections between different pieces of data. In the original it was just about storing courses and being able to search and display them but with the enhancement I had to think about how the courses connected and what could go wrong with the connections.

A challenge I had was making sure the graph represented the prerequisites in the right direction so the program could create a valid course order. I also had to make sure it didn't try to generate an order when circular prerequisites existed so I tested the circular prerequisite connection with some test data and confirmed that the program does detect the cycle and will prevent a course order from being made.